Mustafa Mehboob

Product Engineer · Security · Payments · Cloud

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Summary

Full-stack engineer with **3+ years** building and running revenue-critical web platforms, from database to AWS infrastructure. **Took over a UK visa platform during a live cyber attack**, rebuilt its architecture and **led its three-engineer team as revenue passed $1M**. Delivered products in analytics (LinklyHQ), e-commerce (mDrive Asia), visa processing (WorldVisaHub) and industrial automation (Aeration Control). **First author of a 2026 paper on LLM-based software testing and vulnerability detection**.

Experience

Software Engineer II | **Softaims**, *Lahore, Pakistan* Jan 2026 – Present

UKVisaPortal

- **Took over UKVisaPortal during a live attack**, after a data breach. Contained the incident, traced the entry route and closed the whole class of weakness rather than the single hole. The site stayed up through a Layer-7 DDoS that peaked at 700M requests, ∼**600M of them blocked**, with no data lost.
- **Rebuilt the AWS architecture** around one rule: traffic reaches the application only through CloudFront and the WAF. Locked the origin with CloudFront VPC Origins, moved workloads to ECS Fargate behind a private load balancer, and encrypted passport scans with AES-256-GCM before they are written to S3. Moved the platform off end-of-life Node 20; the image build now refuses an unpatched runtime.
- **Led the three-engineer team** after the recovery, on a TypeScript monorepo with **50+ REST endpoints** over a 39-table PostgreSQL schema: built the admin suite staff use to process applications (call queues, auto-assignment, shift tracking, GDPR erasure and retention), integrated Adyen, MamoPay and PayPal, and built staff sign-in with authenticator-app two-factor and passkeys (WebAuthn). **Revenue passed $1M.**
- **Made agent productivity measurable**: status changes are recorded in the data layer rather than at each call site, with a backfill from the admin audit log, so supervisors see each agent's handle time and hourly throughput.
- **Launched a second brand, Crown Visa,** on the same platform: each brand ships as its own build, which fails if the other brand's text leaks in, and payments, emails and refunds follow the brand and domain a customer used.

WorldVisaHub

- **Co-led WorldVisaHub**, a multi-country visa platform (Rails 8, Inertia, React 19) where supervisors build each country's application form themselves: a versioned form engine with 24 field types, conditional logic and live preview. Made published form versions immutable and pinned to in-flight orders, so an edit never changes an application in progress, and added granular RBAC with per-user overrides and a full admin audit trail.
- **Protected applicant identity documents from credential theft**: one leaked cloud key could have exposed every applicant's documents, so the system holds no permanent keys at all. Uploads go straight to S3 on short-lived STS credentials, files are checked by content rather than extension and encrypted with KMS, and personal data is scrubbed from logs. Added guided camera capture with on-device pose, lighting and blur checks (MediaPipe). Provisioned staging as Infrastructure as Code in Terraform, with keyless OIDC deploys gated on Brakeman and bundler-audit.

Aeration Control

- **Building the NestJS and OPC UA backend of a live industrial SCADA platform** for grain aeration, in which one service holds every site's PLC session: connections open in stages with jittered retries, every node is re-read after an outage, and each organisation's admin manages only its own PLCs. Live values reach the React HMI over SSE through Redis pub/sub, with history in TimescaleDB. Deployed on AWS ECS through Terraform and CI.

Associate Software Engineer | **OptimaGeeks**, *Lahore, Pakistan* Jun 2025 – Dec 2025

- **Led Userbird (now LinklyHQ)**, a GA- and PostHog-style analytics and session-replay platform: an embeddable JavaScript SDK streaming pageviews, events and recordings into BigQuery, with a React and Vite dashboard on Cloudflare Pages. Added caching and lazy loading that made the dashboard **40% faster**. Built tracking from scratch (GTM, GA4, pixels, UTM), plus link shortening, smart redirects and QR codes on LinklyHQ.
- **Shipped production apps** including a LinkedIn Sales Navigator to CRM extension (React, TypeScript), OpenAI-powered CRM tooling, WordPress publishing automation, a bot on Cloudflare Workers and a NestJS backend, deployed with Docker, PM2 and Nginx.

Associate Software Developer | **Webmended** (UAE), *Remote* Mar 2023 – Feb 2025

- **Led mDrive Asia** end to end: storefront, checkout, admin and fulfilment, including NextAuth sign-in, Apple GSX and custom payment integrations (Next.js, Django, PostgreSQL). Also built a Django quality-management system and PHP client projects.

LAMP Stack Developer | **easybew** (Canada), *Remote* Feb 2022 – Mar 2023

Projects

UKVisaPortal | UK visa portal and admin suite 2026
Node.js, TypeScript, PostgreSQL, Prisma, Redis, AWS, Cloudflare

WorldVisaHub | multi-country visa platform with a form builder 2026
Rails 8, Inertia, React 19, PostgreSQL, Sidekiq, AWS

Aeration Control | grain-aeration SCADA 2026
NestJS, OPC UA, Redis, TimescaleDB, React 19, SSE, AWS

LinklyHQ | link shortening, analytics and session replay 2025
React, Vite, BigQuery, Node.js, TypeScript, Cloudflare Pages

mDrive Asia | e-commerce storefront and admin 2023 – 2025
Next.js, NextAuth, Django, PostgreSQL

Mirai | ClickUp-style work management 2025
React, TypeScript, Express, Prisma, PostgreSQL, Redis, Socket.io

Skills

Build: TypeScript, Node.js (NestJS, Express), React, Next.js, Python (Django), Ruby on Rails, OpenAI API, Zustand, Tailwind CSS, Prisma, Socket.io, Sidekiq
Data: PostgreSQL, Redis, MongoDB, BigQuery, TimescaleDB, SQL
Run: AWS (ECS Fargate, RDS, S3, CloudFront, WAF, KMS, IAM/STS), Terraform, Infrastructure as Code (IaC), Docker, GitHub Actions, Cloudflare, Nginx
Integrate: Adyen, MamoPay, PayPal, Zendesk, SendGrid, Slack, Apple GSX, OPC UA, GTM, GA4
Secure: OAuth 2.0, Microsoft SSO, RBAC, TOTP and WebAuthn passkeys, encryption at rest, WAF and DDoS response, GDPR

Publications

First author. "AI-Assisted Automated Software Testing and Vulnerability Detection Using Large Language Models." *Spectrum of Engineering Sciences* 4(3), 2026.
Co-author. "Transfer Learning Optimization for Low-Resource Natural Language Processing Using Transformer Architectures." *Research Consortium Archive* 4(2), 2026.

Education

Government College University (GCU) Lahore Nov 2021 – May 2025
Bachelor of Science in Computer Science (BSCS), CGPA 3.09 *Lahore, Pakistan*